

MAHMOUD AHMED ABDELAWAL

AI & Data Science Engineer | Computer Science Graduate (EELU '26)

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PROFESSIONAL SUMMARY

Results-driven **AI and Data Science Engineer** specializing in machine learning, deep learning, computer vision, and natural language processing. Experienced in architecting production-ready multimodal AI systems, Retrieval-Augmented Generation (RAG) pipelines, disaster detection models, and high-performance FastAPI backends. Demonstrated excellence through winning **1st Place in the Nationwide AI Hackathon** and receiving **Ministerial Recognition from the Minister of Communications and IT of Egypt**. Skilled in Python, PyTorch, TensorFlow, Scikit-Learn, Docker, Azure, and optimized AI inference workflows.

TECHNICAL SKILLS

AI & Machine Learning:	Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Transfer Learning, GANs
Computer Vision:	YOLOv8, Instance Segmentation, Object Detection, CLIP Zero-Shot, BLIP, ConvNeXt, MiDaS Depth Estimation, OpenCV
NLP & RAG:	RAG Pipelines, ChromaDB Vector DB, Nomic Embeddings, Gemini Flash API, SentenceTransformers, Tokenization
Frameworks & Tools:	PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy
Programming:	Python (Expert), Java, C++, JavaScript, TypeScript, SQL
Backend & DevOps:	FastAPI, React 19, SQLite, PostgreSQL, Docker, Microsoft Azure, Git, GitHub, RESTful APIs, VS Code

EDUCATION

Bachelor of Computer Science and Information Technology Egyptian E-Learning University (EELU), Sohag, Egypt Specialization: Artificial Intelligence	Graduation: June 2026
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KEY PROJECTS

S-Report AI: Universal Urban Intelligence & Disaster Management Graduation Project github.com/.../S-Report-AI-Vision
- Developed a hybrid multimodal computer vision system fusing CLIP zero-shot classification with a fine-tuned ConvNeXt-Tiny backbone for sub-150ms disaster identification (fire, flood, collapse, accidents). - Integrated BLIP for automated natural language scene captioning, YOLOv8 for real-time instance segmentation, and MiDaS monocular depth estimation for 3D spatial severity assessment. - Engineered an algorithmic weighted triage decision engine scoring incident urgency for automated emergency resource dispatch. - Architected a production-ready asynchronous FastAPI REST backend with SQLite and containerized Docker deployment.
DermaAssist AI — Bilingual Clinical Decision Support System 1st Place · AI Hackathon github.com/.../AI-Hackathon-Sohag
- Built an award-winning bilingual (Arabic/English) clinical decision support platform strictly grounded in official World Health Organization (WHO) dermatology clinical guidelines. - Implemented an evidence-based RAG pipeline using ChromaDB vector DB, Nomic Embeddings, and Google Gemini Flash with strict clinical prompt safeguards preventing hallucination. - Developed multimodal skin lesion screening (image/video), bilingual voice query interfaces, and a dual-portal doctor/patient electronic health record (EHR) system with automated prescription generation.

- **Tech Stack:** React 19, TypeScript, FastAPI, ChromaDB, SentenceTransformers, SQLite, WebRTC.
- Awarded **1st Place nationwide** at the AI Hackathon sponsored by Ministry of Communications (MCIT), Orange Egypt, and Instant.

Artificial Vincent Van Gogh — Generative Neural Synthesis | DEPI Selected Project

github.com/.../Artificial-Vincent-Van-Gogh

- Architected a generative deep learning pipeline utilizing **DCGANs (Deep Convolutional GANs)** and **Neural Style Transfer (NST)** to synthesize authentic post-impressionist artwork.
- Optimized perceptual, content, and Gram matrix style loss formulations across VGG-19 feature layers using PyTorch and CUDA.

Data Analysis, Statistical Modeling & Insight Extraction | Data Science Portfolio

github.com/.../Data-Analysis

- Conducted rigorous exploratory data analysis (EDA), anomaly detection, missing data imputation, and statistical hypothesis testing (t-test, ANOVA, Chi-Square) across diverse real-world datasets.
- Built reproducible analytical pipelines and high-impact visual dashboards using Python, Pandas, NumPy, Matplotlib, and Seaborn.

ACHIEVEMENTS & HONORS

- ■■■ **Ministerial Recognition for Top Performance:** Formally honored by the **Minister of Communications and Information Technology (MCIT) of Egypt** as one of the top-performing students nationwide within the Digital Egypt Pioneers Initiative (DEPI).
- ■ **1st Place Champion — Nationwide AI Hackathon:** Won first place among engineering teams across Egypt with *DermaAssist AI*, sponsored by the Ministry of Communications & IT, Orange Egypt, and Instant.

CERTIFICATIONS & TRAINING

Microsoft ML Engineer — Digital Egypt Pioneers Initiative (DEPI)	Dec 2025
English & Soft Skills — Digital Egypt Pioneers Initiative (DEPI)	2025
Deep Learning, Computer Vision & NLP — Eng. Hisham Asem Training	2024